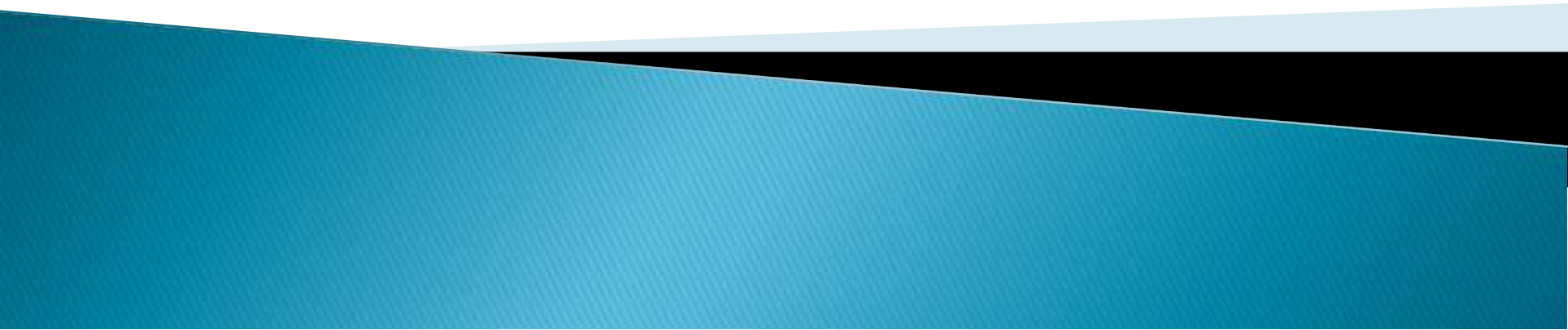


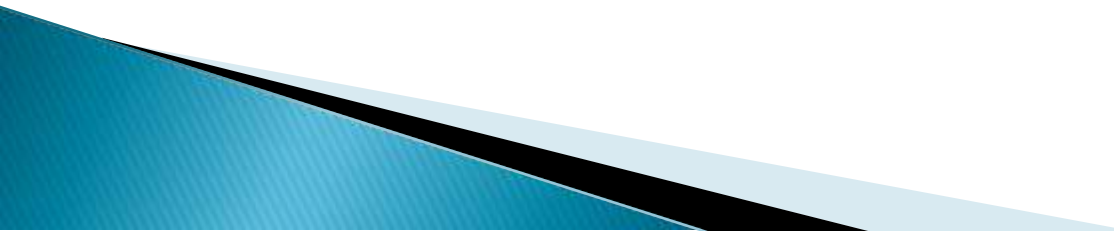
**CAPSTONE PROJECT :
ADVANCED CUSTOMER SEGMENTATION USING
UNSUPERVISED LEARNING**

**PROJECT TITLE :
AI-DRIVEN CUSTOMER INTELLIGENCE SYSTEM FOR
STRATEGIC BUSINESS DECISION MAKING**

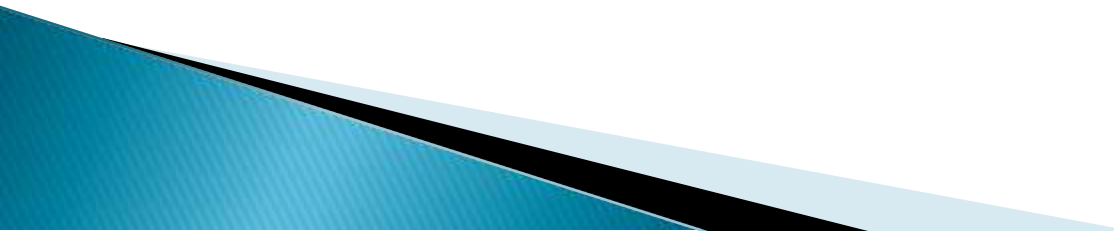
**SUBMITTED BY :
NIVEDITA .D. PURAD
BATCH : AI & ML JAN-2026**



Introduction

- ▶ Customer segmentation is the process of dividing customers into groups based on their characteristics such as demographics, behavior, and spending patterns.
 - ▶ It helps businesses understand customer needs, improve marketing strategies, and increase overall profitability.
 - ▶ With the growth of data, machine learning techniques are widely used to identify hidden patterns in customer data.
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Problem Statement

- ▶ Businesses face challenges in understanding diverse customer behavior.
 - ▶ Without proper segmentation:
 - Marketing strategies become ineffective
 - High-value customers are not identified
 - Customer churn increases
 - ▶ There is a need to automatically group customers and extract meaningful insights from data.
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Objectives

- ▶ The main objectives of this project are:
 - To segment customers using clustering techniques
 - To compare multiple unsupervised algorithms
 - To identify high-value and at-risk customers
 - To provide business insights for decision-making

Dataset Description

- ▶ The dataset contains customer information such as:
 - Age and Income Level
 - Gender and Education Level
 - Occupation and Geographic Information
 - Customer Service Interactions
 - Insurance Products Owned
- ▶ These features help analyze customer behavior and financial capability.

Approach

- ▶ The project follows a complete machine learning pipeline:
 - Data Preprocessing
 - Exploratory Data Analysis (EDA)
 - Feature Engineering
 - Feature Scaling
 - Clustering Model Implementation
 - Model Evaluation and Comparison

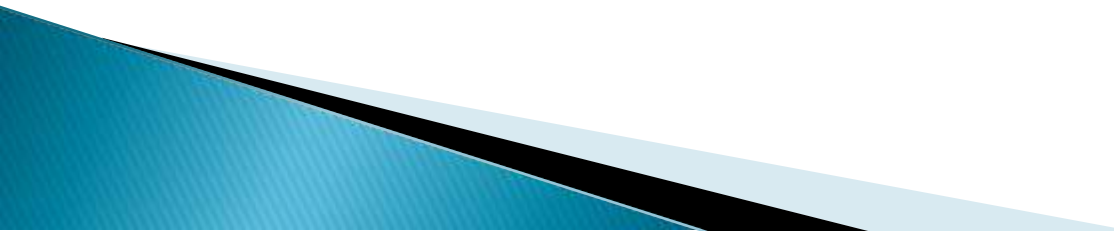
Algorithms Used

- ▶ The following clustering algorithms were implemented:
 - K-Means Clustering
 - Hierarchical Clustering
 - DBSCAN
 - Gaussian Mixture Model (GMM)
- ▶ Each algorithm was analyzed and compared based on performance.

Cluster Optimization

- ▶ To determine the optimal number of clusters:
 - Elbow Method was used to find cluster bend
 - Silhouette Score measured cluster quality
 - Davies-Bouldin Index evaluated cluster compactness
- ▶ Optimal number of clusters selected: $K = 2$

Visualization

- ▶ Principal Component Analysis (PCA) was applied to reduce dimensions and visualize clusters.
 - ▶ The scatter plot shows how customers are grouped into distinct clusters based on similarities.
 - ▶ This helps in understanding cluster separation clearly.
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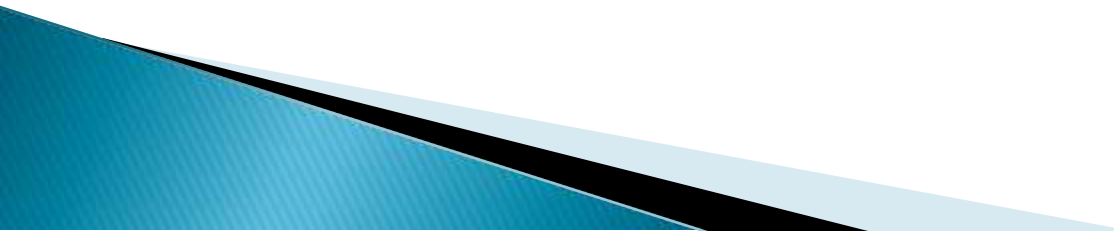
Results & Comparison

- ▶ Performance of algorithms:
 - K-Means: Best performance
 - GMM: Similar performance
 - Hierarchical: Moderate performance
 - DBSCAN: Poor performance
- ▶ K-Means was selected as the best algorithm.

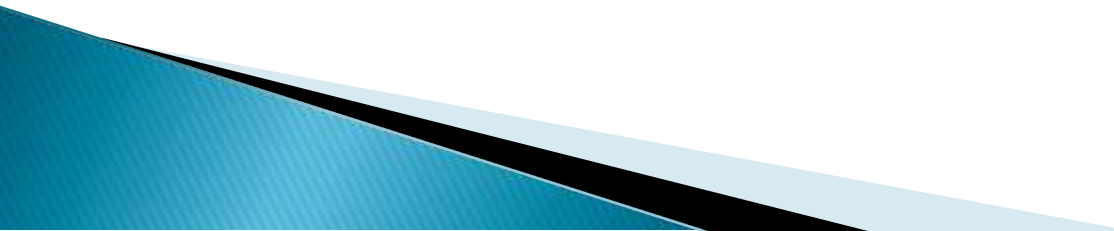
Cluster Explanation

- ▶ Cluster 0:
 - High income customers
 - Stable behavior
 - High value segment
- ▶ Cluster 1:
 - Low income customers
 - Higher service interaction
 - At-risk segment

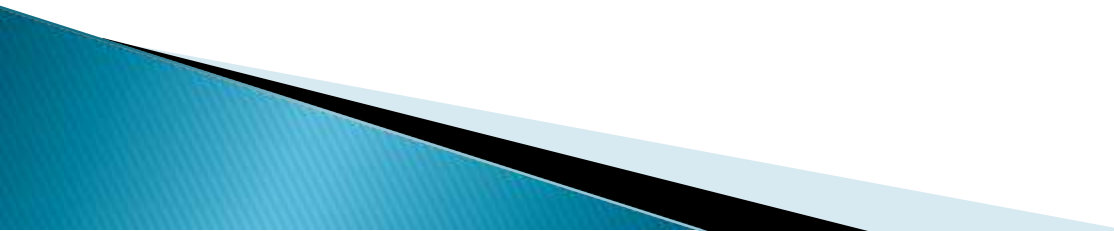
Business Insights

- ▶ High-value customers generate maximum revenue
 - ▶ At-risk customers require retention strategies
 - ▶ Active customers can be targeted for upselling
 - ▶ Personalized marketing can improve engagement
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Advanced Techniques

- ▶ Feature Importance analysis identified key factors like income and interaction
 - ▶ Customer Lifetime Value (CLV) was calculated to identify valuable customers
 - ▶ These techniques enhance business decision-making.
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Conclusion

- ▶ This project demonstrates how unsupervised learning can effectively segment customers.
 - ▶ K-Means clustering provided the best results.
 - ▶ The insights obtained can help businesses improve marketing strategies, increase retention, and enhance profitability.
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THANK YOU